

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		D		1		1		1
		(ii)		$[k = 0.17 \times 1\,000 =] 170 \text{ [N/m]}$		1		1	1	1
	(b)	(i)		$F = kx = 120 \times (10.7 \times 10^{-3}) = 1.284 \text{ [N]} (1)$ Substitution into: $E = \frac{1}{2} Fx$ i.e. $E = 0.5 \times 1.284 \text{ ecf} \times (10.7 \times 10^{-3}) \text{ ecf} (1)$ $= 0.0068694 \text{ [J]} (1)$ Answer of 6.9×10^n where n is not -3 award a maximum of 2 marks	1	1 1		3	3	3
		(ii)		Substitution into $PE = mgh$ i.e. $0.0068694 \text{ ecf} = 0.015 \times 10 \times h (1)$ $h = \frac{0.0068694}{0.015 \times 10} (1\text{-rearrangement})$ $h = 0.045796 \text{ [m]} (1)$ Accept 0.0458 or 0.046 or 0.05 Answer of 4.6×10^n where n is not -2 award a maximum of 2 marks Award 3 marks for an answer of 8.56 with ecf applied for $E = 1.284 \text{ [J]}$ Answer of 8.56×10^n where n is not 0 award a maximum of 2 marks	1	1 1		3	3	3

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		(iii)	I	Dist = area under graph = $0.5 \times 0.9 \times 0.09$ (1) = 0.0405 [m] which is different (1) Accept 0.041 [m]			2	2	2	2
			II	Some of the stored energy in the spring is lost to heat or work done (1) As the spring retracts or friction or air resistance (1)	2			2		2
			III	Choose a spring that has a bigger stiffness constant or stiffer spring / bigger [maximum] extension or longer spring / smaller <u>mass or weight</u> ball / use spring D or F		1		1		1
				Question 5 total	4	7	2	13	9	13